Hadi Khalaf

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Education

Harvard University 2024 – Expected 2028
Ph.D. in Computer Science

- Designing reliable and safe AI systems under weak supervision
- Supported by the Harvard Graduate Prize Fellowship and the Kempner Institute Graduate Fellowship
- *Committee:* Flavio du Pin Calmon (primary advisor), Boaz Barak, and Ariel Procaccia

Harvard University 2024 – 2026
SM in Computer Science, GPA: 3.97/4.0

- *Selected coursework:* reinforcement learning, high-dimensional statistics, nonasymptotic statistics (MIT), optimization (MIT)

American University of Beirut 2020 – 2024
BS in Statistics, BE in Electrical & Computer Engineering with High Distinction, GPA: 4.19/4.0

- Graduated as valedictorian of the Faculty of Engineering and Architecture based on merit and character
- Supported by the AUB Presidential Merit Scholarship (0.4% acceptance)

Publications and Projects *(* indicates first authorship with equal contribution)*

Robust AI Evaluation through Maximal Lotteries ICML 2026
H. Khalaf, Serena Wang, Daniel Halpern, Itai Shapira, Flavio Calmon, Ariel Procaccia

Inference-Time Reward Hacking in Large Language Models NeurIPS 2025 (Spotlight)
H. Khalaf, Claudio Mayrink Verdun, Alex Oesterling, Hima Lakkaraju, Flavio Calmon

AI Alignment at Your Discretion ACM FAccT 2025, Best Paper at NENLP
Maarten Buyl*, **H. Khalaf***, Claudio Mayrink Verdun*, Lucas Monteiro Paes*, Caio Vieira Machado, Flavio Calmon

Experience

Microsoft June 2026 – August 2026
Incoming Applied Scientist Intern

- Building automated evaluation frameworks for enterprise-grade agentic workflows

Beirut Urban Lab Sep 2023 – May 2024
Machine Learning Engineer

- Built and automated an end-to-end agentic pipeline for data elicitation, storage, and retrieval
- Deployed the first GPT-powered tool to crowdsource and query geospatial data over WhatsApp and ArcGIS
- Boosted user engagement with surveys by 72% and user retention by 48%

American University of Beirut Feb 2023 – May 2024
Research Assistant

- Formulated an implicit solution for the information bottleneck problem with non-Gaussian distributions
- Tested new notions of correlation measures for heavy-tailed stochastic processes

Harvard University Jun 2023 – Sep 2023
Research Fellow, Economics Department

- Helped develop the first Bayesian partial identification algorithms for two-player games
- Implemented game-theoretic algorithms to solve two-player games in underspecified economic setups

Selected Awards, Talks, and Scholarships

Kempner Institute Graduate Fellowship 2026

Harvard Berkman Klein Student Leader in AI 2026

Invited Talk at Harvard CS 2902 PhD Seminar 2026

NeurIPS Scholar Award 2025

Princeton Machine Learning Theory Summer School Fellowship 2025

ICML Hudson River Trading Travel Award 2025

Best Paper Award at the New England NLP Workshop at Yale 2025

Invited Talk at the New England NLP Workshop at Yale 2025

Harvard AI Safety Team Technical Fellowship 2024

Harvard Graduate Prize Fellowship 2024
Awarded to select incoming students across Harvard Graduate School of Arts and Sciences based on intellectual promise

AUB High Distinction Award 2024

AUB Penrose Award in the Faculty of Engineering and Architecture 2024
Valedictorian, awarded to the most outstanding graduating student on the basis of scholarship and character

AUB Presidential Scholarship 2020
Full four-year scholarship awarded to the six most outstanding applicants across AUB (0.4% acceptance)

Research Science Institute Fellow at MIT 2019
Hosted as a researcher in the Nuclear Science and Engineering Department at MIT (2% acceptance)

Nationwide Winner in the Lebanese Physics Olympiad 2019
Ranked first over all 11th graders in Lebanon, could not compete at IPhO due to lack of governmental funding